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#include "esp_camera.h"
#include <WiFi.h>
#include <WiFiClientSecure.h>

// WiFi Credentials
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// Telegram Bot Details
String BOTtoken = "YOUR_BOT_TOKEN";
String chat_id = "YOUR_CHAT_ID";

WiFiClientSecure client;

#define PIR_PIN 13

// AI Thinker Camera Pin Configuration
#define PWDN_GPIO_NUM 32
#define RESET_GPIO_NUM -1
#define XCLK_GPIO_NUM 0
#define SIOD_GPIO_NUM 26
#define SIOC_GPIO_NUM 27
#define Y9_GPIO_NUM 35
#define Y8_GPIO_NUM 34
#define Y7_GPIO_NUM 39
#define Y6_GPIO_NUM 36
#define Y5_GPIO_NUM 21
#define Y4_GPIO_NUM 19
#define Y3_GPIO_NUM 18
#define Y2_GPIO_NUM 5
#define VSYNC_GPIO_NUM 25
#define HREF_GPIO_NUM 23
#define PCLK_GPIO_NUM 22

void startCamera() {
    camera_config_t config;
    config.ledc_channel = LEDC_CHANNEL_0;
    config.ledc_timer = LEDC_TIMER_0;
    config.pin_d0 = Y2_GPIO_NUM;
    config.pin_d1 = Y3_GPIO_NUM;
    config.pin_d2 = Y4_GPIO_NUM;
    config.pin_d3 = Y5_GPIO_NUM;
    config.pin_d4 = Y6_GPIO_NUM;

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config.pin_d5 = Y7_GPIO_NUM;
config.pin_d6 = Y8_GPIO_NUM;
config.pin_d7 = Y9_GPIO_NUM;
config.pin_xclk = XCLK_GPIO_NUM;
config.pin_pclk = PCLK_GPIO_NUM;
config.pin_vsync = VSYNC_GPIO_NUM;
config.pin_href = HREF_GPIO_NUM;
config.pin_sscb_sda = SIOD_GPIO_NUM;
config.pin_sscb_scl = SIOC_GPIO_NUM;
config.pin_pwdn = PWDN_GPIO_NUM;
config.pin_reset = RESET_GPIO_NUM;
config.xclk_freq_hz = 20000000;
config.pixel_format = PIXFORMAT_JPEG;
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config.frame_size = FRAMESIZE_VGA;
config.jpeg_quality = 10;
config.fb_count = 1;
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esp_camera_init(&config);
}
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void sendPhotoTelegram() {
    camera_fb_t * fb = NULL;
    fb = esp_camera_fb_get();
    if (!fb) {
        Serial.println("Camera capture failed");
        return;
    }
}
```

```
client.setInsecure();
if (!client.connect("api.telegram.org", 443)) {
    Serial.println("Connection failed");
    return;
}
```

```
String head = "--RandomNerdTutorials\r\nContent-Disposition: form-data;
name=\"chat_id\";\r\n\r\n" + chat_id +
    "\r\n--RandomNerdTutorials\r\nContent-Disposition: form-data; name=\"photo\";
filename=\"image.jpg\";\r\nContent-Type: image/jpeg\r\n\r\n";
String tail = "\r\n--RandomNerdTutorials--\r\n";
```

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uint16_t imageLen = fb->len;
uint16_t extraLen = head.length() + tail.length();
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uint16_t totalLen = imageLen + extraLen;

client.println("POST /bot" + BOTtoken + "/sendPhoto HTTP/1.1");
client.println("Host: api.telegram.org");
client.println("Content-Length: " + String(totalLen));
client.println("Content-Type: multipart/form-data; boundary=RandomNerdTutorials");
client.println();

client.print(head);

client.write(fb->buf, fb->len);
client.print(tail);

esp_camera_fb_return(fb);
Serial.println("Photo sent");
}

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("WiFi connected");

  startCamera();
}

void loop() {
  int motion = digitalRead(PIR_PIN);

  if (motion == HIGH) {
    Serial.println("Motion detected!");
    sendPhotoTelegram();
    delay(10000); // prevent multiple triggers
  }
}

```